Alan Lyu

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Summary

I am Alan Lyu, an undergraduate student majoring in Physics at Peking University while also pursuing a second degree in Life Sciences. My research interests are biophysics and computational neuroscience. I use tools and frameworks from physics to study how complex structures and functions emerge in living systems. I hope to pursue doctoral study in biophysics, physics, or computational neuroscience.

Education

Peking University, Bachelor of Science — *Sep. 2023 – Jul. 2027 (expected)*

- **Major:** Physics
- **Second Degree:** Life Science
- **CGPA:** 3.97/4.00*

  ** Note: Beginning in fall 2025, the School of Physics at Peking University no longer provides official GPA or cohort ranking. This GPA was calculated from the original grades of individual courses using the WES conversion method.*

- **Related Coursework:** 3.99 in Thermal Physics; 3.91 in Equilibrium Statistical Physics; 3.95 in Methods for Mathematical Physics; 3.97 in Electrodynamics; 3.93 in Computational Physics; 3.88 in Topics on Nonlinear Physics; A- in Physical Chemistry in Cellular and Molecular Biology; 3.95 in Lectures on Systems Biology; A in Fundamentals of Network Science and Systems Biology; A in Mathematical Modeling in the Life Sciences.

Research

Adaptive Sessile-Motile Switching in *Bacillus subtilis*

Advisor: Prof. Chao Tang

- Built an agent based model from steady environment measurements to simulate sessile and motile state switching in fluctuating chemical environments.
- Used transfer entropy to compare how agents with bidirectional regulation and one way regulation use environmental information in decision making.
- Evaluated the adaptive value of the two switching strategies during chemotaxis.

Functional Optimization of the *Drosophila* Olfactory Circuit

Advisors: Prof. Thierry Emonet and Dr. Kevin Chen

- Extended a circuit for odor motion sensing into a parameterized network that can be optimized.
- Designed a joint task for odor motion and odor gradient sensing under a metabolic constraint and optimized network parameters for task performance.
- Analyzed the tradeoff between the two sensory tasks within a shared network.

Emergence of Low Rank Structure in Artificial Neural Networks

Advisors: Prof. Yuhai Tu and Dr. Yikuan Zhang

- Trained multilayer perceptrons on MNIST and recorded representations and singular value spectra throughout learning.
- Studied the stages of low rank structure formation by varying network width and learning rate and by regularizing intermediate singular values.
- Examined whether the observed scale dependence admits a useful renormalization description.

Teaching

Teaching Assistant (Dry Lab), Cell Biology Laboratory *Fall 2026*

School of Life Sciences, Peking University

- Will guide undergraduate students in applying computer vision algorithms to identify stages of the cell cycle in HeLa cells labeled by immunofluorescence and in root tip cells of *Vicia faba* stained by the Feulgen method.

Honors and Awards

- National Scholarship of China & Peking University Merit Student Award 2024–2025
- National Scholarship of China & Peking University Merit Student Award 2023–2024
- Finalist in the Mathematical Contest in Modeling (MCM) 2026
- Meritorious in the Mathematical Contest in Modeling (MCM) 2025
- Peking University Undergraduate Physicists' Tournament (PKUPT) 2023–2024
  - Second Prize (Team rank 2/18)
  - Best Opponent and Best Reviewer Awards